

Практикум A2

77 marks of the archive, seven techniques, thirteen tasks. Everything the archive asks about geometric sequences from November 2021 to November 2025 — including the halves of questions that A1 had to leave behind, because the exam likes to start a question arithmetic and finish it geometric.

О практикуме

Собран из двух тем сразу:

`number_algebra.geometric_sequences` целиком (17 блоков, 55 баллов) и остаток `number_algebra.sequences` после A1 (10 блоков, 22 балла) — те вопросы, что начинаются арифметической последовательностью и кончаются геометрической.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-a2-geometric-sequences

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

A2 — Geometric sequences and infinite sums

77 marks of the archive, seven techniques, thirteen tasks. Everything the archive asks about geometric sequences from November 2021 to November 2025 — including the halves of questions that A1 had to leave behind, because the exam likes to start a question arithmetic and finish it geometric.

The one idea

A geometric sequence differs from an arithmetic one by a single word.

From each term to the next you multiply by the same thing.

$$u_n = u_1 r^{n-1}$$

is "*you have multiplied by r exactly $n - 1$ times*", and the -1 is the same count of steps that it was in A1, lost just as often.

The sum comes out of one trick. Write S_n , write rS_n underneath, and subtract: everything in the middle cancels.

$$\begin{array}{rcl} S_n & = & u_1 + u_1 r + u_1 r^2 + \cdots + u_1 r^{n-1} \\ rS_n & = & u_1 r + u_1 r^2 + \cdots + u_1 r^{n-1} + u_1 r^n \end{array}$$

$$S_n - rS_n = u_1 - u_1 r^n$$

$\implies$

$$S_n = \frac{u_1 (r^n - 1)}{r - 1}$$

The other idea, which arithmetic sequences do not have

If $|r| < 1$ the terms shrink, the tail can be made smaller than any number you name, and the **infinite** series has a sum:

$$S_\infty = \frac{u_1}{1 - r} \quad \text{when } |r| < 1$$

If $|r| \geq 1$ it has none. That is not a footnote. Half the questions in this topic are about which of the two you are in, and "*state a reason why the sum of an infinite number of terms does not exist*" is a question the exam asks in those words.

And a third, which is where the marks go

The ratio is found by **dividing**, and dividing gives a **quadratic** whenever the unknown sits inside the terms. A quadratic has two roots, and something else in the question chooses between them: a sign, a convergence condition, an index that has to be whole.

$r^2 = 9$ has two answers. Writing $r = 3$ costs a mark, by name: the markscheme awards *M1A1A0* for $\sqrt{\cdot}$ with *no* $\pm$.

How the checks work

They do not know the answers, and they do not know the formulas. Each check is handed **the rule** — a first term and a ratio — and multiplies:

```
verify_term('1 (r)', 0.74088, geometric(80, qlr), 4)
```

is *"your ratio: is the fourth term of that sequence 0.74088?"* The term is reached

by multiplying three times. $u_1 r^{n-1}$ appears nowhere.

The infinite sum is the same idea taken as far as it goes: `verify_infinite` adds term after term until the tail stops mattering. So a series that does not shrink cannot be summed by it at all — and when you hand it one, it says so in the examiner's own words.

The checks run in both directions, as they did in A1: forward when the sequence is known and the answer is a term or a sum, backward when the answer is inside `geometric(...)` and the number in the line came from the question.

When you are wrong the check says **how**:

what you wrote	what the check says
$u_1 r^n$	that is the next term: the index is out by one
u_1/u_2	the ratio is upside down
3 where -3 was meant	the sign is gone: an alternating sequence has a negative ratio
S_∞ where S_n was asked	that is the sum of the whole sequence, not of the first n
$u_1/(1+r)$	the denominator is $1+r$ instead of $1-r$
n one too small	at that index the condition does not hold yet

Order of work

level	what it means	tasks
	the sequence is given; walk it	1-4
	the ratio is what you are looking for	5-7
	the ratio is not a number, or n is the unknown	8-13

Every task is a real past-paper question, cited.

52% of these marks are on a calculator paper, and for once the

number is close to honest. The smallest- n questions are genuinely solved by a table of values — the markscheme lists *table*, *sketch* and *logarithms* side by side — and the November 2025 particle needs a GDC to find the three maxima it then asks you to divide.

```

import sys
sys.path.append('.')          # from
practicum/number_algebra to practicum/kit/
import sympy as sp           # the escape
hatch: anything not in kit is in sp
from kit import *             # checks +
geometric, term, total, infinite, ...

language('en')                # this
notebook is in English, and so are the checks

n = symbols('n')              # the index:
every one of these papers calls it n
m = symbols('m')              # a second
index, for the timer
r = symbols('r')              # the ratio,
when it is the unknown
a, s, t = symbols('a s t')    # letters that
stay letters
x = symbols('x')              # the ratio is
allowed to be an expression in x
                                # k already
comes from kit

# Every sequence below may be built from an
answer you have not written
# yet. geometric() returns an empty rule in
that case, so the notebook
# runs top to bottom while it is still blank
and the checks print .

def ratios(rule):
    # the geometric sequence whose nth term
    is rule(n)
    try:

```

```

        first, second = rule(1), rule(2)
    except TypeError:                                     #
an answer inside the rule is blank
        return geometric(..., ...)
    if blank(first, second) or first == 0:
        return geometric(..., ...)
    return geometric(first, second / first)

print('ready; sympy', sp.__version__)
print('a geometric sequence:', geometric(3,
2))
print('its 5th term:          ',
term(geometric(3, 2), 5))
print('S5:                    ',
total(geometric(3, 2), 5))
# the infinite sum is reached by adding, so
it arrives as a long fraction
print('and all of 1,  $\frac{1}{2}$ ,  $\frac{1}{4}$ , ...:',
float(infinite(geometric(1, Rational(1,
2)))))

```

Map of the seven techniques

#	technique	you recognise it by	it reduces to
1	the ratio and a term	two terms given, or a percentage	$u_n = u_1 r^{n-1}$
2	add the first n	<i>sum</i> , a Σ with a top limit	$S_n = \frac{u_1(r^n - 1)}{r - 1}$
3	the infinite sum	S_∞ , <i>converges</i> , <i>forever</i>	$\frac{u_1}{1-r}$, and $ r < 1$ first
4	a series written as Σ	no term is written out at all	read off u_1 and r
5	three quantities in ratio	<i>form a geometric sequence</i> , a letter inside	$s^2 = at$
6	the smallest n	<i>smallest</i> , <i>least</i> , <i>how many years</i>	logs, or a table
7	the ratio is an expression	$1 + x + x^2 + \dots$, a Σ with x inside	the same formulas, in letters

Techniques 1–3 are the three formulas, and 3 carries the condition that makes it legal. Technique 4 is reading, not computing: the work is seeing u_1 and r in a notation that hides them. Technique 5 is where the topic stops being arithmetic in disguise — dividing produces a

quadratic, and quadratics have two answers. Technique 6 is the exam's favourite way of forcing an integer out of a continuous answer, and technique 7 is everything above with r that is not a number.

Part 1. Walking the sequence

Theory: the ratio, and how many times it is applied

Take the sequence 11, 44, 176, 704, ... Each term is four times the one before, so $r = 4$, and

$$u_5 = 11 \cdot 4^4 = 2816$$

Four multiplications, not five: between u_1 and u_5 there are four steps. The exponent is the **number of steps**, and everything in this part is that sentence applied twice.

The ratio is found by dividing, and the division may skip terms. If you are given $u_2 = 44$ and $u_5 = 2816$, then between them there are three steps:

$$u_5 = u_2 r^3 \implies r^3 = \frac{2816}{44} = 64 \implies r = 4$$

Odd root, one answer. Even root, two. $r^3 = 64$ gives $r = 4$ and nothing else. But $r^2 = 9$ gives $r = 3$ **and** $r = -3$,

and the question that follows almost always wants the second one.

Percentages are ratios wearing a coat. *Increases by 3%* is $r = 1.03$; *decreases by 40%* is $r = 0.6$. The only difficulty is the numbering: if u_1 is the value after the first year, then the value after n years is $u_n = u_1 r^{n-1}$, and writing r^n counts the first year twice.

Task 1 — **May 2025 TZ2 Paper 2**
Q7(a), 3 marks

A geometric sequence has first term 80 and fourth term 0.74088.

(a) Find the second term.

Write the ratio down as well as the term. A wrong ratio and a wrong second term are two different mistakes, and it is worth knowing which one you made.

```
qlr = ...          # the common ratio
qlu = ...          # the second term

mine = geometric(80, qlr)
# the first term is given

verify_term('1 (r)', 0.74088, mine, 4)
# is the 4th term of yours 0.74088?
verify_term('1a', qlu, mine, 2)
```

Theory: adding the first n , and counting them

The subtraction at the top of this notebook gives

$$S_n = \frac{u_1 (r^n - 1)}{r - 1} = \frac{u_1 (1 - r^n)}{1 - r}$$

Those are the same formula: multiply top and bottom of one by -1 and you get the other. Use whichever keeps the numbers positive — with $r > 1$ the first, with $r < 1$ the second — but never mix the top of one with the bottom of the other.

For $11 + 44 + 176 + 704$ that is $u_1 = 11$, $r = 4$, four terms:

$$S_4 = \frac{11 (4^4 - 1)}{4 - 1} = \frac{11 \cdot 255}{3} = 935$$

Count the terms, do not assume them. The number of terms is what the formula's exponent is, and it is not always the last index you can see. In $1 + x + x^2 + \cdots + x^n$ there are $n + 1$ terms, because the first one is x^0 . There is a mark for saying so.

A sum of sums is still a sum. If $S_1 + S_2 + \cdots + S_n$ is asked, write each S_k out first: it usually splits into a geometric series plus something trivial, and the geometric half is what this formula is for.

Task 2 ● — May 2024 TZ1 Paper 1
Q5, 5 marks

Consider a geometric sequence with first term 1 and common ratio 10. S_n is the sum of the first n terms of the sequence.

(a) Find an expression for S_n in the form $\frac{a^n - 1}{b}$, where $a, b \in \mathbb{Z}^+$.

(b) Hence, show that $S_1 + S_2 + S_3 + \cdots + S_n = \frac{10(10^n - 1) - 9n}{81}$.

Part (b) is a "show that", so there is nothing to hand over at the end of it. What is asked instead is the step that carries the marks: after writing $S_k = \frac{10^k - 1}{9}$, the sum splits into $\frac{1}{9} \sum 10^k$ minus $\frac{1}{9} \sum 1$, and only the first half is geometric. **Write down that first half**, $10 + 100 + \dots + 10^n$, in terms of n .

```
q2s = ...          # S_n, in terms of n
q2t = ...          # the value of 10 + 100 +
1000 + ... + 10**n, in terms of n

mine = geometric(1, 10)

verify_total('2a', q2s, mine, n)
verify_total('2b', q2t, geometric(10, 10), n)
# the same ratio, ten times bigger
```

Task 3 ● — May 2025 TZ1 Paper 1 Q5, 8 marks

Consider a sequence of ten rectangular picture frames $F_1, F_2, \dots, F_{10}$. Frame F_1 has width 4 cm and height 5 cm. The width and height of frame F_n are each increased by 50% to give the width and height of F_{n+1} , for $1 \leq n \leq 9$.

(a) (i) Show that the area of frame F_n is $20 \left(\frac{9}{4}\right)^{n-1} \text{ cm}^2$. **(ii)** Hence find the **mean** area of the ten frames, in the form $p \left(\left(\frac{9}{4}\right)^a - 1\right) \text{ cm}^2$, where $p \in \mathbb{Q}^+$, $a \in \mathbb{Z}^+$. **(b)** Find the **median** area of the ten frames, in the form $q \left(\frac{9}{4}\right)^4 \text{ cm}^2$, where $q \in \mathbb{Q}^+$.

Both sides grow by the same factor, so the area grows by its square — that is where $\frac{9}{4}$ comes from, and (a)(i) is that sentence.

Ten areas have no middle one: the median sits between the fifth and the sixth. The check uses that directly — your median, doubled and added to the first four areas, has to be the sum of the first six.

```

q3p = ...          # p
q3a = ...          # a
q3q = ...          # q

areas = geometric(20, Rational(9, 4))
# from (a)(i)

# ten of your means make the total of ten
areas
from_mean = ... if blank(q3p, q3a) else 10 *
q3p * (Rational(9, 4) ** q3a - 1)
# the first four areas plus twice your median
make the first six
from_median = (... if blank(q3q)
                else total(areas, 4) + 2 * q3q
* Rational(9, 4) ** 4)

verify_total('3a(ii)', from_mean, areas, 10)
verify_total('3b', from_median, areas, 6)

```

Theory: adding all of them, when that is allowed

Let n grow in $S_n = \frac{u_1(1 - r^n)}{1 - r}$. If $|r| < 1$ then $r^n \rightarrow 0$, the bracket goes to 1, and

$$S_\infty = \frac{u_1}{1 - r}$$

If $|r| \geq 1$ the terms do not shrink and there is no such number. Both halves of

that are examinable, and the second one more often than the first.

$$45 + 18 + 7.2 + \dots$$

$$r = 0.4,$$

$$S_{\infty} = \frac{45}{1 - 0.4} = 75$$

The condition is $|r| < 1$, not $r < 1$.

With $r = -3$ the terms are $7, -21, 63, \dots$ — they grow, and -3 is certainly less than one. Writing the condition without the modulus signs is the standard way of losing that mark.

"Show that the series converges" is answered by naming $|r|$ and comparing it with 1 — in one line, for **both** values of r if the question produced two. Leaving out the second one is R0.

Total distance is not the sum of the distances. A particle that swings out to u_1 and back to the start has travelled $2u_1$ in that interval. If the question asks how far it travels altogether, the sequence to sum is $2u_1, 2u_2, \dots$

Task 4 ● — November 2021 Paper 2
Q5, 9 marks

The sum of the first n terms of a geometric sequence is given by

$$S_n = \sum_{i=1}^n \frac{2}{3} \left(\frac{7}{8} \right)^i$$

(a) Find the first term of the sequence, u_1 . **(b)** Find S_∞ . **(c)** Find the least value of n such that $S_\infty - S_n < 0.001$.

The Σ starts at $i = 1$, so the first term is the whole summand at $i = 1$ — not the fraction standing in front of the bracket. That is technique 4 arriving early, and it is worth two marks here.

In (c) the check walks the sequence and stops at the first n where your own S_∞ satisfies the condition — so a wrong (b) costs you (b) only.

```

q4u = ...           # the first term
q4s = ...           # the sum to infinity,
exactly
q4n = ...           # the least n with  $S_{\infty} - S_n < 0.001$ 

mine = geometric(q4u, Rational(7, 8))
# the ratio is visible in the sigma

verify_term('4a', Rational(2, 3) *
Rational(7, 8) ** 2, mine, 2)
verify_infinite('4b', q4s, mine, exact=True)
verify_least('4c', q4n, mine, lambda partial:
q4s - partial < Rational(1, 1000))

```

Part 2. When the ratio is what you are looking for

Theory: three quantities in a geometric sequence

" a, s, t form a geometric sequence" is not a list. It is one equation, and it says the two ratios are equal:

$$\frac{s}{a} = \frac{t}{s} \quad \implies \quad s^2 = at$$

The middle term squared is the product of its neighbours — the same statement as " s is the geometric mean of a and t ". Whenever three things are said to be in geometric sequence, write that line first; everything else in the question comes out of it.

With a letter inside, it becomes a quadratic. For 11, w , 44:

$$w^2 = 11 \cdot 44 = 484 \implies w = \pm 22$$

and both are real sequences: 11, 22, 44 with $r = 2$, and 11, -22 , 44 with $r = -2$.

Sometimes it collapses to a linear equation instead, and that is not a mistake — it is the quadratic terms cancelling. For $h - 1$, $h + 1$, $h + 5$:

$$\begin{aligned}(h + 1)^2 &= (h - 1)(h + 5) \\&\implies h^2 + 2h + 1 \\&= h^2 + 4h - 5 \\&\implies h \\&= 3\end{aligned}$$

giving 2, 4, 8. Expand both sides before deciding what kind of equation you have.

The difference is not the ratio.

Equating $u_2 - u_1$ with $u_3 - u_2$ tests for an *arithmetic* sequence. The exam puts both conditions in one question precisely to see which one you write.

Task 5 — **May 2024 TZ2 Paper 1 Q10(b), 2 marks**

Consider the geometric sequence $a, s, t \dots$, where $a, s, t \neq 0$.

(b) Show that $s^2 = at$.

A "show that" with nothing to hand over — so write the third term instead. Given a and s , what must t be?

```
q5t = ...           # the third term, in terms
of a and s

verify_geometric('5', [a, s, q5t])
```

Task 6 — **May 2023 TZ1 Paper 1 Q10 (d)-(e), 5 marks**

(Parts (a)–(c) of this question are arithmetic and are in A1. They establish that the sequence u_n has $u_1 = 5$ and $u_6 = 15$.)

Consider a geometric sequence v_n , where $v_2 = u_1$ and $v_4 = u_6$.

(d) Find the possible values of the common ratio, r . **(e)** Given that $v_{99} < 0$, find v_5 .

The check for (d) is built out of the sequence itself: it takes your r , walks two steps from v_2 , and asks whether it lands on 15 — then solves that same condition itself to see whether you have missed one.

```

q6r = [...]      # both possible values of
the common ratio
q6n = ...        # the one that makes v99
negative
q6v = ...        # v5

# v2 is the first term the question gives us,
so this sequence starts there:
# its 3rd term is v4.
verify_root_set('6d', q6r,
Eq(term(geometric(5, r), 3), 15), var=r)
verify_term('6e', q6v, geometric(5, q6n), 4)

```

Task 7 — *May 2025 TZ3 Paper 1* **Q10(b), 4 marks**

Consider the sequence $\{u_n\}$ whose first three terms are

$$\begin{aligned}
 u_1 &= k - 5, \\
 u_2 &= 3 - 2k, \\
 u_3 &= 5k + 3, \\
 k &\in \mathbb{R}
 \end{aligned}$$

Consider the case where $k = 12$.

(i) Show that the first three terms of $\{u_n\}$ form a geometric sequence. **(ii)** Given that $\{u_n\}$ is geometric, state a reason why the sum of an infinite number of terms of this sequence does not exist.

Part (ii) has no cell: it is one sentence, and the sentence is in the solutions. For (i), write the three terms out and the ratio you got by dividing them — the check compares your terms with the question's own expressions at $k = 12$, and separately asks whether they are geometric.

```
q7t = [...]      # the first three terms when
k = 12
q7r = ...        # the common ratio

printed = [k - 5, 3 - 2 * k, 5 * k + 3]
# the question's own expressions

verify_start('7(i)', q7t, ratios(lambda i:
printed[i - 1].subs(k, 12)))
verify_geometric('7(i) geometric', q7t)
verify_ratio('7(i) r', q7r, q7t)
```

● Part 3. The smallest n , and ratios that are not numbers

Theory: forcing an integer out

"Find the least value of n such that ..." is never solved by solving. The inequality gives a fractional boundary, and the answer is the integer on the correct side of it.

$$\begin{aligned}2 \cdot 3^n > 5000 &\implies 3^n > 2500 \\&\implies n > \frac{\ln 2500}{\ln 3} \\&= 7.122\dots \\&\implies n \\&= 8\end{aligned}$$

Three ways to lose this:

- **rounding the wrong way.** $n > 7.122$ rounds *up*. Had the inequality been $<$, it would round *down*, and the boundary itself would need checking;
- **taking logs of a shrinking ratio without flipping.** $\ln r < 0$ when $|r| < 1$,

and dividing by it reverses the inequality;

- **using u_n where the question said S_n** . The November 2021 markscheme awards **M0** for that by name.

On a calculator paper you do not have to take logs at all. The markscheme lists *sketch* **OR** *table of values* **OR** *algebraic manipulation involving logarithms* as equals. A table is often faster, and it never flips a sign the wrong way.

Check the neighbour. Whatever route you took, evaluate at n and at $n - 1$. Two numbers, ten seconds, and the answer is either confirmed or off by one — which is the only way this question is ever wrong.

Task 8 — **November 2022 Paper 2 Q3, 5 marks**

A geometric sequence has a first term of 50 and a fourth term of 86.4. The sum of the first n terms of the sequence is S_n .

Find the smallest value of n such that $S_n > 33\,500$.

```
q8r = ...           # the common ratio
q8n = ...           # the smallest n with S_n >
33500

mine = geometric(50, q8r)

verify_term('8 (r)', 86.4, mine, 4)
verify_least('8', q8n, mine, lambda partial:
partial > 33500)
```

Task 9 — **May 2024 TZ1 Paper 2**
Q1, 7 marks

Darren buys a car for \$35 000. The value of the car decreases by 15% in the first year.

(a) Find the value of the car at the end of the first year.

After the first year, the value decreases by 11% in each subsequent year.

(b) Find the value of Darren's car 10 years after he buys it, giving your answer to the nearest dollar.

(c) When Darren has owned the car for n complete years, its value is less than 10%

of its original value. Find the least value of n .

*The first year is different from all the others, so it is not part of the geometric sequence — it produces the sequence's **first term**. After that the ratio is 0.89 and the value after n years is the n th term, not the $(n + 1)$ th.*

```
q9a = ...          # the value after one year
q9b = ...          # the value after ten years,
to the nearest dollar
q9n = ...          # the least number of
complete years

mine = geometric(q9a, 0.89)
# your first term, the stated ratio

verify_term('9a', 35000 * 0.85, mine, 1)
verify_term('9b', q9b, mine, 10)
verify_least('9c', q9n, mine, lambda value:
value < 3500, what='term')
```

Theory: reading a series that was never written out

A Σ hides both numbers you need, and neither is where it looks.

$$\sum_{i=1}^{\infty} \left(\frac{2}{5}\right)^i = \frac{2}{5} + \frac{4}{25} + \frac{8}{125} + \dots$$

- **the first term** is the whole summand evaluated at the *lower limit* — here $\frac{2}{5}$, and if the limit had been $i = 0$ it would be 1;
- **the ratio** is what changes from one summand to the next, which is whatever is being raised to the power — here $\frac{2}{5}$ again, but only by coincidence.

$$S_{\infty} = \frac{2/5}{1 - 2/5} = \frac{2}{3}$$

A constant in front of the bracket is **not** the first term. In $\sum_{i=1}^{\infty} c \rho^i$ the first term is $c\rho$, and reading it as c is worth two marks in task 4.

Repeating decimals are geometric series, and that is the whole technique behind writing them as fractions. $0.\overline{4}$ is $\frac{4}{10} + \frac{4}{100} + \dots$ with $r = \frac{1}{10}$, giving $\frac{4/10}{9/10} = \frac{4}{9}$. When only part of the decimal repeats, split it: the non-repeating head is a separate fraction, added on at the end.

Task 10 ● — **November 2025 TZ1**
Paper 2 Q6, 5 marks

Consider the infinite series

$$\sum_{k=0}^{\infty} \left(\frac{53}{1000} \right) \left(\frac{1}{100} \right)^k.$$

(a) Find the exact value of S_{∞} .

Let $0.2\overline{53}$ represent the repeating decimal $0.2535353\dots$

(b) Use your answer from part (a) to express $0.2\overline{53}$ as a fraction whose numerator and denominator have no common factors.

*This Σ starts at $k=0$, so here the constant in front **is** the first term. Compare with task 4, where it was not.*

In (b) the check subtracts the head of the decimal from your fraction and asks whether what is left is the series — so the two answers are tied together, and (b) cannot accidentally be right if (a) is wrong.

```

q10s = ...          # the exact sum to infinity
q10f = ...          # 0.2535353... as a fraction
in lowest terms

series = geometric(Rational(53, 1000),
Rational(1, 100))

verify_infinite('10a', q10s, series,
exact=True)
verify_infinite('10b', ... if blank(q10f)
else q10f - Rational(1, 5),
series, exact=True)

```

Theory: when the ratio is an expression

Nothing in the three formulas needs r to be a number. In

$$1 + \frac{x}{3} + \frac{x^2}{9} + \dots$$

the ratio is $\frac{x}{3}$, and everything goes through unchanged:

$$\begin{aligned}
 S_n &= \frac{1 - (x/3)^n}{1 - x/3}, & S_\infty \\
 &= \frac{1}{1 - x/3} \\
 &= \frac{3}{3 - x} \text{ for } |x| < 3
 \end{aligned}$$

Two things change, and both carry marks.

The convergence condition becomes a set of x . $|r| < 1$ is an inequality in x once you substitute, and it is x the question asks about: $|\frac{x}{3}| < 1$ means $-3 < x < 3$.

The excluded value has to be named. S_n divides by $r - 1$, so $r = 1$ is not allowed, and a question about $1 + x + x^2 + \dots + x^n$ says "*for $x \neq 1$* " for that reason. Saying it back is part of the answer.

Watch the sign inside the ratio. For $1 - x^2 + x^4 - \dots$ the ratio is $-x^2$, not x^2 , and $1 - (-x^2)$ is $1 + x^2$. Half the marks in this technique are that one minus sign.

Task 11 — **November 2022 Paper 3 Q1(e), 2 marks**

By considering $f(x) = 1 + x + x^2 + \dots + x^n$ as a geometric series, for $x \neq 1$, show that

$$f(x) = \frac{x^{n+1} - 1}{x - 1}$$

*Two marks, and one of them is **R1** for a clear indication of how many terms there are. Write that number down first.*

```
q11n = ...      # how many terms the series
                 has, in terms of n
q11s = ...      # the sum of that many terms

verify_total('11', q11s, geometric(1, x),
             q11n, var=x, values=(2, 3, Rational(1, 2)))
```

Task 12 — **May 2025 TZ1 Paper 2 Q12(b) and November 2025 TZ3 Paper 3 Q2(a), 4 marks**

(i) Consider $f_n(x) = \sum_{r=0}^n (-2x^2)^r$. Given that $f_3(x) = 1 - 2x^2 + 4x^4 - 8x^6$, write down a similar expression for $f_4(x)$ in ascending powers of x .

(ii) Given $|x| < 1$, find the sum to infinity of the geometric series $1 - x^2 + x^4 - x^6 + \dots$

In (i) the ratio is $-2x^2$ and the sum is finite — count the terms from $r = 0$. In (ii) it is $-x^2$ and the sum is infinite, so the answer is a function of x .

```

q12f = ...          # f4(x), in ascending powers
of x
q12s = ...          # the sum to infinity of 1 -
x^2 + x^4 - ...

verify_total('12(i)', q12f, geometric(1, -2 *
x ** 2), 5,
                var=x, values=(2, Rational(1,
3)))
verify_infinite('12(ii)', q12s, geometric(1,
-x ** 2),
                var=x, values=(Rational(1,
2), Rational(1, 3)))

```

Task 13 — November 2025 TZ3 Paper 2 Q10(e), 5 marks

A particle P moves in a straight line so that its displacement from a fixed point O at time t is $s(t) = 2^{(1-\frac{t}{5})} \sin(\frac{2\pi t}{3})$, $t \geq 0$. It passes through O every T seconds.

A sequence $u_1, u_2, u_3 \dots$ is formed where $u_1, u_2, u_3 \dots$ are the largest **distances** from O in each of the intervals $0 < t < T$, $T < t < 2T$, $2T < t < 3T$ respectively. Earlier parts of the question give

$$u_1 = 1.80645,$$

$$u_2 = 1.46729,$$

$$u_3 = 1.19181$$

It is known that $u_1, u_2, u_3 \dots$ form a geometric sequence.

(i) Determine the value of the common ratio r . **(ii)** Calculate the **total distance** travelled by the particle if it were to continue to move in this way indefinitely.

In each interval the particle goes out to its farthest point and comes back, so it covers twice that distance. The check sums a sequence whose first term is $2u_1$ — which is exactly the modelling step the markscheme gives its third mark for.

```
q13r = ...          # the common ratio, to three
significant figures
q13d = ...          # the total distance
travelled

out_and_back = geometric(2 * 1.80645, 1.46729
/ 1.80645)  # out and back each time

verify_ratio('13(i)', q13r, [1.80645,
1.46729, 1.19181])
verify_infinite('13(ii)', q13d, out_and_back)
```

Trainer: name the technique in five seconds

Twelve openings. Do not compute anything — say only **which move you would make first**.

code	technique
ratio	two terms, or a percentage; divide to get r
sum	a finite number of terms is being added
infinite	S_∞ , <i>forever</i> , or <i>why does it not exist</i>
sigma	the series is written only as a Σ
condition	<i>form a geometric sequence</i> , with a letter inside
least	<i>smallest n, how many years</i> ; round an integer out
letter	the ratio is an expression in x

1. A geometric sequence has $u_1 = 6$ and $u_4 = 48$. Find u_2 .
2. Find $9 + 3 + 1 + \frac{1}{3} + \dots$ continued forever.
3. The numbers p , $p + 4$ and $4p$ form a geometric sequence. Find p .
4. A colony doubles every day and starts at 200. After how many days does it

pass a million?

5. Evaluate $\sum_{i=1}^{\infty} \frac{7}{4} \left(\frac{1}{3}\right)^i$.

6. Find $2 + 6 + 18 + \cdots + 2 \cdot 3^9$.

7. For which x does $1 + 4x + 16x^2 + \cdots$ have a sum?

8. A ball rebounds to 60% of its height each bounce. Explain why the total distance is finite.

9. The terms $u_1 = m - 1$, $u_2 = m + 3$, $u_3 = 5m - 3$ are geometric. Find m .

0. A savings account grows by 4% a year from \$500. When does it first exceed \$800?

1. A sequence has $u_3 = 20$ and $u_6 = 2.5$. Find the common ratio.

2. Write $\sum_{k=0}^{\infty} \left(\frac{x}{2}\right)^k$ as a rational function.

```
answers = {
    1: '', 2: '', 3: '', 4: '', 5: '', 6: '',
    7: '', 8: '', 9: '', 10: '', 11: '', 12:
    '',
}

trigger_check(answers, {1: '2f04a79ef13b', 2:
'704b7acbacf9', 3: '596d40a4ed24', 4:
'6bab65a2bdde', 5: '38de90475bb3', 6:
'09f5ffef2830', 7: 'f80105efab6d', 8:
'704b7acbacf9', 9: '596d40a4ed24', 10:
'6bab65a2bdde', 11: '2f04a79ef13b', 12:
'f80105efab6d'})
```

On the clock — *May 2025 TZ3* *Paper 1 Q10(c), 7 marks*

Seven marks, ten minutes. No hints this time.

The sequence $\{u_n\}$ has first three terms

$$u_1 = k - 5,$$

$$u_2 = 3 - 2k,$$

$$u_3 = 5k + 3,$$

$$k \in \mathbb{R}$$

The sequence is geometric for a second value of k (the first, $k = 12$, was task 7).

(i) Show that $k^2 - 10k - 24 = 0$. **(ii)** Find the first three terms of $\{u_n\}$ for this second value of k . **(iii)** Hence, write down the value of S_{2m} , the sum of the first $2m$ terms, for this second value of k .

Attempt log

date	time	result

```
qt_k = ...          # the second value of k
qt_r = ...          # the common ratio it gives
qt_t = [...]        # the first three terms
qt_s = ...          # S_2m

printed = [k - 5, 3 - 2 * k, 5 * k + 3]
# the question's own expressions

verify_root_set('timer (i)', [12, qt_k], Eq(k
** 2 - 10 * k - 24, 0), var=k)
verify_geometric('timer (ii)', qt_t)
verify_start('timer (ii) terms', qt_t,
              ratios(lambda i: printed[i -
1]).subs(k, qt_k)
                  if not blank(qt_k) else
...))
verify_total('timer (iii)', qt_s,
geometric(qt_t[0], qt_r), 2 * m)
```

Solutions

Work these only after you have your own answer, or you are reading, not practising.

1 From u_1 to u_4 there are three steps:

$$\begin{aligned}80r^3 &= 0.74088 \\ \implies r^3 &= 0.0092610 \\ \implies \boxed{r = 0.21}\end{aligned}$$

The cube root is exact — $0.21^3 = 0.009261$ — which is the paper telling you the ratio is a tidy number and you have not slipped. Then

$$u_2 = 80(0.21) = \boxed{16.8}$$

2 (a) $u_1 = 1$, $r = 10$, so

$$S_n = \frac{1(10^n - 1)}{10 - 1} = \boxed{\frac{10^n - 1}{9}}$$

that is $a = 10$, $b = 9$.

2 (b) Sum those:

$$\sum_{k=1}^n S_k = \sum_{k=1}^n \frac{10^k - 1}{9} = \frac{1}{9} \left(\sum_{k=1}^n 10^k - \sum_{k=1}^n 1 \right)$$

The first sum is geometric with first term 10, ratio 10 and n terms:

$$\sum_{k=1}^n 10^k = \frac{10(10^n - 1)}{9}$$

and the second is just n . So

$$\frac{1}{9} \left(\frac{10(10^n - 1)}{9} - n \right) = \frac{10(10^n - 1) - 9n}{81}$$

The two nines multiplying into 81 is the whole point of the given form; if your denominator is 9, you have divided once instead of twice.

3 (a)(i) Width and height are both geometric with ratio $\frac{3}{2}$, so

$$\text{area of } F_n = 4 \left(\frac{3}{2}\right)^{n-1} \times 5 \left(\frac{3}{2}\right)^{n-1} = 20 \left(\frac{9}{4}\right)^{n-1}$$

(ii) The areas are geometric with $u_1 = 20$,
 $r = \frac{9}{4}$:

$$S_{10} = \frac{20 \left(\left(\frac{9}{4} \right)^{10} - 1 \right)}{\frac{9}{4} - 1} = 16 \left(\left(\frac{9}{4} \right)^{10} - 1 \right)$$

and the mean is a tenth of that:

$$\frac{16}{10} \left(\left(\frac{9}{4} \right)^{10} - 1 \right) = \boxed{\frac{8}{5} \left(\left(\frac{9}{4} \right)^{10} - 1 \right)}$$

so $p = \frac{8}{5}$ and $\boxed{a = 10}$. Forgetting to divide by ten is the standard loss here — the question says *mean*, not *total*.

3 (b) Ten values have no middle one, so the median is the mean of the 5th and the 6th:

$$\begin{aligned} \frac{20 \left(\frac{9}{4} \right)^4 + 20 \left(\frac{9}{4} \right)^5}{2} &= \frac{20 \left(\frac{9}{4} \right)^4 \left(1 + \frac{9}{4} \right)}{2} \\ &= \boxed{\frac{65}{2} \left(\frac{9}{4} \right)^4} \end{aligned}$$

Factoring $\left(\frac{9}{4} \right)^4$ out of both terms is what turns the answer into the demanded form; expanding instead gives a correct number in the wrong shape, and the shape is what is marked.

4 (a) The summand at $i = 1$:

$$u_1 = \frac{2}{3} \cdot \frac{7}{8} = \frac{14}{24} = \boxed{\frac{7}{12}}$$

Not $\frac{2}{3}$. The $\frac{7}{8}$ is inside the sum, so it belongs to every term including the first.

4 (b) $r = \frac{7}{8}$, and $|\frac{7}{8}| < 1$:

$$S_\infty = \frac{7/12}{1 - 7/8} = \frac{7/12}{1/8} = \boxed{\frac{14}{3}}$$

4 (c) What is left after n terms is itself a geometric series, so

$$S_\infty - S_n = \frac{14}{3} - \frac{\frac{7}{12} (1 - (\frac{7}{8})^n)}{\frac{1}{8}} < 0.001$$

Solving — by table, sketch or logarithms — gives $n > 63.267\dots$, so

$$\boxed{n = 64}$$

Check the neighbours, which is what the markscheme prints: $S_\infty - S_{63} = 0.001036$ (too big) and $S_\infty - S_{64} = 0.000906$ (small enough). Solving $S_\infty - u_n < 0.001$ instead of $S_\infty - S_n$ scores **M0** — the difference between a term and a sum, again.

5 Equal ratios:

$$\frac{s}{a} = \frac{t}{s} \implies s^2 = at \implies \boxed{t = \frac{s^2}{a}}$$

Two marks, and the second is for a correct equation before the rearrangement — $\left(\frac{s}{a}\right)^2 = \frac{t}{a}$ counts, $s^2 = at$ written straight down does not.

6 (d) $v_2 = 5$ and $v_4 = 15$ are two steps apart:

$$\begin{aligned} v_4 &= v_2 r^2 \\ &\implies 5r^2 \\ &= 15 \\ &\implies r^2 \\ &= 3 \\ &\implies \boxed{r = \pm\sqrt{3}} \end{aligned}$$

The markscheme is explicit: $\sqrt{3}$ alone, with no working, is **M1A1A0**.

6 (e) $v_{99} = v_2 r^{97}$, and 97 is odd, so the sign of v_{99} is the sign of r . Negative means $r = -\sqrt{3}$, and then

$$\begin{aligned}
 v_5 &= v_2 r^3 \\
 &= 5 \left(-\sqrt{3} \right)^3 \\
 &= 5 \left(-3\sqrt{3} \right) \\
 &= \boxed{-15\sqrt{3}}
 \end{aligned}$$

which is $-\frac{45}{\sqrt{3}}$ in the markscheme's form — the same number.

7 (i) With $k = 12$ the terms are

$$\begin{aligned}
 u_1 &= 7, & u_2 &= -21, & u_3 &= 63 \\
 \frac{-21}{7} &= -3, & \frac{63}{-21} &= -3
 \end{aligned}$$

Equal, so the three terms are geometric with $\boxed{r = -3}$.

7 (ii) $|r| = 3 \geq 1$, so the terms do not shrink and the infinite sum does not exist. Saying " $r < 1$ is false" is not enough on its own: the condition is on $|r|$, and $-3 < 1$ is true.

8 From u_1 to u_4 , three steps:

$$50r^3 = 86.4 \implies r^3 = 1.728 \implies r = 1.2$$

$$\begin{aligned} S_n &= \frac{50(1.2^n - 1)}{0.2} \\ &= 250(1.2^n - 1) > 33\,500 \end{aligned}$$

$$\begin{aligned} 1.2^n > 135 &\implies n > \frac{\ln 135}{\ln 1.2} \\ &= 26.9045\dots \\ &\implies \boxed{n = 27} \end{aligned}$$

The neighbours: $S_{26} = 28\,368.8$ and $S_{27} = 34\,092.6$. Rounding 26.9 down is the only mistake available here, and it is the common one.

9 (a) A 15% loss leaves 85%:

$$0.85 \times 35\,000 = \boxed{\$29\,750}$$

9 (b) From then on the ratio is 0.89, and 29 750 is the value after **one** year, so ten years later is nine more steps:

$$29\,750 \times 0.89^9 = \boxed{\$10\,423}$$

Using 0.89^{10} counts the first year twice and gives \$9276.

9 (c) 10% of the original is \$3500:

$$29\,750 \times 0.89^{n-1} < 3500 \implies n > 19.364\dots$$

$$\implies \boxed{n = 20}$$

At $n = 19$ the car is worth \$3651.80 — still too much — and at $n = 20$ it is \$3250.10. The finance app on a GDC answers the whole thing with $I\% = -11$, $PV = -29750$, $FV = 3500$.

10 (a) This Σ starts at $k = 0$, so the first term is the constant itself and the ratio is $\frac{1}{100}$:

$$S_{\infty} = \frac{53/1000}{1 - 1/100} = \frac{53}{1000} \cdot \frac{100}{99} = \boxed{\frac{53}{990}}$$

which is $0.0535353\dots$ — the repeating tail of the decimal in (b).

10 (b) Split the decimal at the point where repetition starts:

$$\begin{aligned}
0.2535353 \dots &= 0.2 + 0.0535353 \dots \\
&= \frac{1}{5} + \frac{53}{990} \\
&= \frac{198}{990} + \frac{53}{990} \\
&= \boxed{\frac{251}{990}}
\end{aligned}$$

251 is prime, so nothing cancels. Trying to write the whole decimal as a single series instead — with $\frac{253}{1000}$ as a first term — does not work: the 2 does not repeat.

11 The series runs $x^0, x^1, \dots, x^n$, so there are $\boxed{n+1}$ terms, with $u_1 = 1$ and $r = x$:

$$f(x) = \frac{1(x^{n+1} - 1)}{x - 1} = \boxed{\frac{x^{n+1} - 1}{x - 1}}$$

The **R1** is for the count. Writing $\frac{x^n - 1}{x - 1}$ — the formula applied with n terms — is the mistake the mark exists to catch.

12 (i) f_4 has one more term than f_3 , and the next power of $-2x^2$ is $(-2x^2)^4 = 16x^8$:

$$f_4(x) = \boxed{1 - 2x^2 + 4x^4 - 8x^6 + 16x^8}$$

Five terms, because r runs from 0 to 4.

12 (ii) Here $u_1 = 1$ and $r = -x^2$, and $|x| < 1$ makes $|r| < 1$:

$$S_\infty = \frac{1}{1 - (-x^2)} = \boxed{\frac{1}{1 + x^2}}$$

The sign is the whole question. Reading the ratio as x^2 gives $\frac{1}{1-x^2}$, which is the sum of a different series.

13 (i) Divide any two consecutive terms:

$$r = \frac{1.46729}{1.80645} = 0.812252 \dots \implies \boxed{r = 0.812}$$

Dividing u_3 by u_1 gives r^2 , not r — take the square root if you go that way.

13 (ii) Each interval takes the particle out to u_i and back, so the distance covered in it is $2u_i$:

$$\begin{aligned}
\text{total} &= 2 \sum_{i=1}^{\infty} u_i \\
&= 2 \cdot \frac{1.80645}{1 - 0.812252} \\
&= 2(9.62167) \\
&= \boxed{19.2 \text{ cm}}
\end{aligned}$$

The markscheme gives a mark for the doubling on its own, and warns against any answer smaller than $2u_1 = 3.61$ — which is what you get if the ratio came out wrong.

Timer

(i) Geometric means the two ratios are equal:

$$\begin{aligned}
\frac{3 - 2k}{k - 5} &= \frac{5k + 3}{3 - 2k} \\
&\implies (3 - 2k)^2 \\
&= (k - 5)(5k + 3)
\end{aligned}$$

$$\begin{aligned}
9 - 12k + 4k^2 &= 5k^2 - 22k - 15 \\
&\implies k^2 - 10k - 24 \\
&= 0
\end{aligned}$$

(ii) $(k - 12)(k + 2) = 0$, and $k = 12$ was task 7, so $\boxed{k = -2}$. The terms are

$$u_1 = -2 - 5 = -7,$$

$$u_2 = 3 + 4 = 7,$$

$$u_3 = -10 + 3 = -7$$

$$\boxed{-7, 7, -7}$$

with $r = -1$. Substituting k into the wrong expression is the standard slip; u_3 is $5k + 3$, not $3 - 2k$.

(iii) With $r = -1$ the terms cancel in pairs:

$$(-7 + 7) + (-7 + 7) + \dots$$

and $2m$ terms make exactly m such pairs, so

$$\boxed{S_{2m} = 0}$$

No formula is needed, and using one is worse than useless: $r = -1$ makes $\frac{u_1(r^{2m}-1)}{r-1}$ a fraction $\frac{0}{-2}$, which is right but tells you nothing about why.